

2 (**long**) Consider the amplifier circuit in Fig. 2.

(a) At a mid-band frequency, where C_1 may be considered to be an open circuit, calculate the value of R_2 required to give a voltage gain of 300 between input and output. The operational amplifier may be considered ideal. [8]

(b) What is the mid-band input impedance of the circuit? [6]

(c) Calculate the value of C_1 required to give a 3 dB high frequency cut-off at 10 kHz, where the circuit gain drops to $1/\sqrt{2}$ of its mid-band value. [6]

(d) In practice, the operational amplifier has an open-loop voltage gain of 20 000, but is otherwise ideal. What is the actual mid-band voltage gain in dB when the circuit drives a load impedance of $25\ \Omega$? [10]

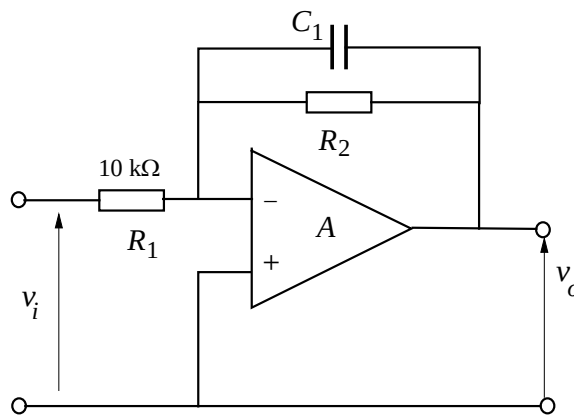


Fig. 2